

Ultrahigh-Throughput Activity Engineering of Promiscuous Amidases through a Fluorescence-Activated Cell Sorting Assay

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Cite This: *ACS Catal.* 2025, 15, 8902–8912



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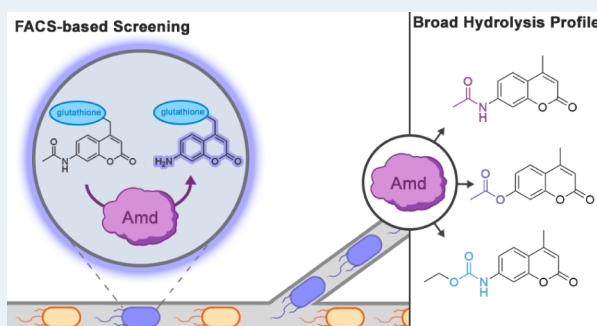
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ABSTRACT: Ultrahigh-throughput methods such as flow cytometry are ideal tools for the directed evolution of enzymes by enabling the screening of up to 10^9 variants per day. In this study, we developed an assay based on fluorescence-activated cell sorting (FACS) for the detection and engineering of amidase activity in whole cells. The assay establishes a stable genotype–phenotype linkage by coupling coumarin-based hydrolysis products to intracellular glutathione via a recombinantly coexpressed glutathione S-transferase. To demonstrate the applicability of the FACS-based assay, we engineered an amidase from *Sphingomonas alpine* (SaAmd) by screening combinatorial libraries with multiple amino acid positions randomized simultaneously. SaAmd variants containing proximal double mutations exhibited not only almost 5-fold improved activity against structurally different amide substrates but also coevolved promiscuous carbamate- and ester-hydrolyzing activities, which exceeded the wildtype activity up to 6-fold. Importantly, triple variants featuring distal mutations in three highly flexible loop regions, displayed up to 16-fold enhanced specific activities toward small molecules containing highly stable *N*-aryl amide and carbamate bonds. These motifs are commonly used as protecting groups for amines in organic synthesis but can also be found in environmental contaminants like pesticides and plastic waste. Therefore, the developed FACS-assisted assay has great potential to accelerate the engineering of amidases for versatile biotechnological applications.

KEYWORDS: amidase, biocatalysis, flow cytometry, high-throughput screening, loop engineering

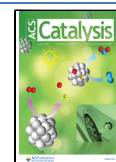


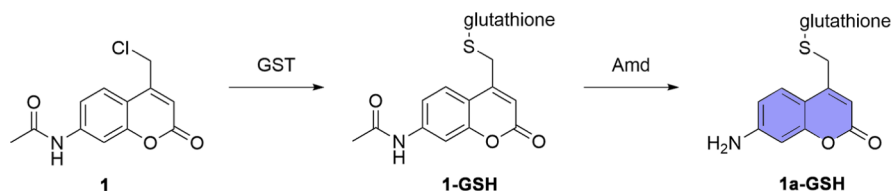
INTRODUCTION

Biocatalysts have evolved to complement – and even substitute – established chemical catalysts in synthetic schemes, yielding value-added pharmaceuticals, as well as fine and bulk chemicals that are essential to different industries.^{1,2} Driven by recent advancements in the *omics* fields, DNA technologies, and continuous improvement of bioinformatic tools, there is a rapidly growing number of functionally annotated enzyme-coding sequences, which can be further customized through protein engineering.¹ Additionally, the recent introduction of machine learning-based tools to reliably predict protein structures from their primary sequence and suggest beneficial mutations, for example, greatly accelerated both the *de novo* design and the directed evolution of enzymes.^{1,3} Together, these technological advancements facilitated the tailoring of biocatalysts to install “new-to-nature” activities or to meet the requirements for industrial application through increasingly complex enzyme engineering campaigns.^{1,3–7} While one to five mutations per protein scaffold were typical in the early 2000s, these advancements have increased the number of feasible amino acid substitutions to over 40 in a single enzyme.^{8–10} Larger engineering campaigns

often employ randomized or combinatorial libraries – constructed by error-prone PCR (epPCR)^{11–13} or (iterative) saturation mutagenesis^{14,15} of multiple sites, respectively – that can contain numbers of variants exceeding the capacity for detailed biochemical characterization. Especially simultaneous site-saturation mutagenesis (SSM) of multiple residues can unlock combinations of amino acid exchanges that exhibit epistatic effects.^{16–18} Since epistasis in protein engineering reflects non-additive effects resulting from interactions of specific mutations, the corresponding variants are difficult to predict and can be easily overlooked by iterative combination of point mutations.^{19,20} When aiming at the simultaneous randomization of five residues with oligonucleotides containing degenerate NNK codons, for example, more than 10^7 variants

Published: May 12, 2025



Scheme 1. Reaction Scheme for the Detection of Amidase Activity through the FACS-Based Assay^a

^aConjugation of AACMC (1) and GSH by GST yields the intermediate 1-GSH, which is retained within cells. Upon amide bond cleavage by an amidase, the fluorescent target product 1a-GSH is formed, which allows the sorting of cells exhibiting improved amidase activity, as 1 and 1-GSH are non-fluorescent.

have to be screened – excluding the oversampling required to ensure 99% library coverage.^{21,22} Hence, efficient ways to navigate this vast sequence space are necessary.

In this regard, ultrahigh-throughput screening (uHTS) strategies, including flow cytometry-based screening and selection assays, like fluorescence-activated cell sorting (FACS), are highly desirable.^{23–25} FACS devices are capable of screening up to 10⁸ clones per day, enabling the analysis of enzyme variants containing a large number of amino acid exchanges.^{23,26} An important consideration for uHTS methods is a stable link between phenotype and genotype. The phenotype is exhibited by a protein, for example, an enzyme with improved catalytic activity, while the genotype is the nucleic acid sequence that encodes it.²⁷ For the engineering of enzymatic activities through the FACS of whole cells, it is essential that the activity reporter is retained within the cell and unable to diffuse out of it as described below.²⁸ Complementary, emerging technologies like microfluidics utilize different *in vitro* compartmentalization strategies (e.g., generation of water-in-oil-in-water double emulsion droplets) to both link genotype and phenotype and be compatible with FACS setups.^{4,27,29} Considering these requirements, the advantages of flow cytometry-based screening systems have successfully been implemented for a multitude of directed evolution campaigns.^{28,30–33}

In this study, we aimed at exploiting the inherent ultrahigh throughput of FACS for the engineering of amidases. Amidases comprise several protein families, such as the amidase signature (AS) superfamily (InterPro: IPR036928), the nitrilase family (InterPro: IPR023919), and the acetamidase/formamidase family (InterPro: IPR054833).³⁴ Generally, amidases can perform amide bond cleavage reactions on a broad range of substrates, with excellent regio- and enantioselectivity. They operate independently of cofactors and yield an amine and a carboxylic acid as hydrolysis products (Figure S1).³⁵ Hence, members of different amidase families have been applied for the production of enantio-pure carboxylic acids,^{36,37} chiral amino acids,^{38,39} as well as pharmaceutically active compounds, including acetohydroxamic acid^{40,41} and fatty hydroxamic acids.⁴² Only recently, amidases, particularly members of the AS family, have moved into the spotlight due to their capability to hydrolyze highly stable *N*-aryl carbamate and amide bonds, present in synthetic polymers like polyurethanes and polyamides, for example.^{43,44} Consequently, tools for the rapid engineering of polymer-degrading enzymes and the reliable assessment of related hydrolytic activities are urgently needed. In this regard, an absorbance-based HT assay in a microtiter plate format was developed and successfully applied for the directed evolution of enzymes with polyamidase activity.⁴⁵ Although other assays for the detection of amidase activity have been established in the past, they are either

limited by insufficient screening throughput (~10³ variants per day)^{46–48} or based on highly specialized selection linked to cell metabolism.^{49–51} Considering these shortcomings, we envisioned a flow cytometry-based screening, using the coumarin-derived amide substrate 7-acetamido-4-chloromethylcoumarin (AACMC, 1; Scheme 1). Different coumarins have been used in fluorescence-based screenings, including FACS.^{28,32,48,52} To retain the fluorophore within amidase-expressing cells and guarantee a stable phenotype-genotype linkage, we utilized the electrophilic carbon of the 4-chloromethyl group to be conjugated to intracellular glutathione (GSH) through the activity of a heterologously co-expressed glutathione *S*-transferase (GST).^{53–55} The corresponding intermediate product (1-GSH) is trapped within cells. Upon cleavage of the acetamide group, the fluorescent product 7-amino-4-methylcoumarin-GSH (1a-GSH) is formed (Scheme 1).^{53,54} The reaction can take place in reverse order as well, yielding the same fluorescent reaction product (for a full reaction scheme, see Scheme S1). This assay principle was challenged by the screening of large libraries of an uncharacterized amidase from *Sphingomonas alpina* (SaAmd). Libraries were created through SSM, randomizing multiple amino acid residues in SaAmd, proximal and distal to each other. Following the successful FACS-assisted assay establishment, our fluorescence-based uHTS setup enabled the screening of combinatorial amidase libraries with a throughput of 500 events per second (>10⁸ variants per day). Remarkably, not only did the screening improve amidase activity >16-fold but coevolved the promiscuous esterase and urethanase activity of SaAmd as well.

MATERIALS AND METHODS

General Information. Chemicals were obtained from Carl Roth, Merck, Honeywell Fluka, Sigma-Aldrich, or Thermo Fisher Scientific and used without further purification unless stated otherwise.

Genes encoding enzymes investigated in this work were codon-optimized for the expression in *Escherichia coli* (*E. coli*), synthesized, and sub-cloned into pET-26b(+) vectors – in-frame with a backbone-encoded C-terminal 6xHis-tag – by BioCat GmbH (Heidelberg, Germany). Protein sequences are given in the Supporting Information. The following strains were ordered from Thermo Fisher Scientific: *E. coli* BL21-(DE3)-Gold and *E. coli* TOP10. Chemo-competent cells were prepared using established protocols and transformed by heat shock.⁵⁶ The SaAmd structure model was created using AlphaFold2.⁵⁷

Cloning and Library Construction. Gibson assembly was used to create pACYCDuet-1_*gst*, using the Gibson Assembly Master Mix from New England Biolabs (Frankfurt am Main,

Germany) and following the instructions of the supplier. Both the *gst* gene and the pACYCDuet-1 backbone⁵⁸ were amplified by polymerase chain reaction (PCR) as described in the SI. For the creation of NNK libraries of *SaAmd*, the QuikChange method was adapted for SSM as described previously.⁴³ The QuikChange primer design tool by Agilent was used to design the oligonucleotides containing the desired NNK codons (<https://www.agilent.com/store/primerDesignProgram.jsp>). Desalted oligonucleotides were ordered from Eurofins Genomics GmbH (Ebersberg, Germany). Successful assembly and mutagenesis were confirmed by Sanger sequencing by Microsynth AG (Balgach, Switzerland), using the company's standard primers (T7, T7term) unless stated otherwise. All primers are listed in Table S1.

FACS-Based Assay. Pre-cultures of *E. coli* BL21(DE3)-Gold cells, harboring pACYCDuet-1-*gst* and the desired pET-26b(+)-encoded amidase, were grown in 5 mL lysogeny broth (LB medium), supplemented with 25 $\mu\text{g}\cdot\text{mL}^{-1}$ chloramphenicol and 50 $\mu\text{g}\cdot\text{mL}^{-1}$ kanamycin, at 37 °C in an orbital shaker (140 rpm) overnight. For enzyme production, a main culture was prepared by inoculating 5 mL LB medium supplemented with both antibiotics with 1% (v/v) of pre-culture. It was cultivated as before. Once the optical density measured at 600 nm (OD_{600}) reached 0.5–0.7, isopropyl β -D-thiogalactopyranoside (IPTG) and AACMC (**1**) were added to a final concentration of 1 mM and 0.1 mM, respectively; **1** was added from a stock solution (25 mM) prepared in dimethyl sulfoxide (DMSO). Enzyme production was performed at 20 °C (140 rpm). After 24 h, the final OD_{600} was determined and the cells were harvested by centrifugation (2,400 g, 4 °C, 1 min). The supernatant was discarded and the cell pellet was resuspended in 100 mM potassium phosphate buffer (pH 7.5) for analysis by flow cytometry. Cells were diluted (1:1,000) in 2 mL potassium phosphate buffer in screening cuvettes to achieve event rates of 500 to 3,000 cells per second. Cells were sorted with a gate threshold containing the top 0.5–1% of fluorescent cells. Sorted cells were collected in 50 mL reaction tubes and an equal amount of 2X SOC medium (4% tryptone, 1% (w/v) yeast extract, 20 mM NaCl, 5 mM KCl, 20 mM MgCl_2 , 20 mM MgSO_4 , 40 mM glucose). Cells were recovered at 37 °C (140 rpm) for 2 h and spread onto LB agar plates supplemented with 50 $\mu\text{g}\cdot\text{mL}^{-1}$ kanamycin and 25 $\mu\text{g}\cdot\text{mL}^{-1}$ chloramphenicol. Plates were incubated at 37 °C overnight. Single colonies were picked the following day to inoculate overnight cultures as described above. Plasmid DNA was isolated with the innuPREP Plasmid Mini Kit 2.0 (Analytik Jena, Jena, Germany) as instructed by the supplier and sent for Sanger sequencing. For the enrichment experiment, the sorted cells were used to inoculate an LB culture (supplemented with 50 $\mu\text{g}\cdot\text{mL}^{-1}$ kanamycin and 25 $\mu\text{g}\cdot\text{mL}^{-1}$ chloramphenicol for the selection of cells harboring both an amidase-coding pET26b(+) vector and pACYCDuet-1-*gst*, respectively) and grown overnight. The following day, a main culture was inoculated with the resulting overnight culture, and protein expression was performed as before.

Flow cytometry analysis was conducted with a Partec CyFlow space device (Sysmex Partec GmbH, Görlitz, Germany). The flow cytometer is equipped with an ultraviolet (UV) diode laser (375 nm, 16 mW), a blue solid-state laser (488 nm), and a green solid-state laser (532 nm). Amidase activity was detected using **1**. The GSH conjugates were excited by a UV laser diode and measured with the default setup and filter (FL3-UV: $\lambda_{\text{ex}} = 375/30 \text{ nm}/\lambda_{\text{em}} = 440/30 \text{ nm}$).

Calibration beads were used prior to the screening to ensure the correct excitation and emission.

Trapping of **1**-GSH and fluorescence exhibited by **1a**-GSH after cleavage of the amide bond was also analyzed by fluorescence microscopy (see Figure S3). Images were obtained using a Leica DM2500 LED microscope (Leica Microsystems) equipped with a CoolLED pE-300white (SB) (CoolLED) lighting system ($\lambda_{\text{ex}} = 350/50 \text{ nm}/\lambda_{\text{em}} = 460/50 \text{ nm}$) and analyzed using the LAS X software v. 3.3.3. 16958 (Leica Microsystems).

High-Throughput Protein Production and Purification. Chemically competent *E. coli* BL21(DE3)-Gold cells were transformed with pET-26b(+)-*SaAmd*, a desired variant, or pET-26b(+)-*PdAmd*. Pre-cultures were prepared by inoculating 0.2 mL LB medium supplemented with 50 $\mu\text{g}\cdot\text{mL}^{-1}$ kanamycin with single colonies of the desired transformant in sterile 96-deep well plates. Cultures were grown at 37 °C with shaking (800 rpm) in an orbital plate shaker TiMix TH30 (Edmund Bühler GmbH, Bodelshausen, Germany) overnight. Main cultures were prepared by inoculating 0.8 mL of PSAM-5052 medium⁴³ supplemented with 50 $\mu\text{g}\cdot\text{mL}^{-1}$ kanamycin with 2% (v/v) of the pre-culture. After 5 h of incubation at 37 °C with shaking (800 rpm), the temperature was shifted to 20 °C for enzyme expression. After 16–20 h, cells were harvested by centrifugation (3,000 g, 4 °C, 20 min). The supernatant was discarded and the cell pellets were resuspended in 0.2 mL lysis buffer, containing 0.5X BugBuster (diluted from 10X Protein Extraction Reagent; Merck-Millipore, Darmstadt, Germany) and bovine DNase I (0.1 U·mL⁻¹; Sigma-Aldrich, Taufkirchen, Germany) in 50 mM Tris-HCl, 100 mM NaCl (pH 8.0). The cells were incubated at 25 °C (1,000 rpm) for 10 min for cell lysis, followed by a centrifugation step (3,000 g, 4 °C, 45 min). The cell-free extract (CFE) was transferred onto His MultiTrap FFs 96-well plates (Cytiva Europe GmbH, Freiburg, Germany). Purification was performed as instructed by the manufacturer and as previously described.⁴³ The protein concentration was assessed using 10 μL of eluates with the PierceTM BCA Protein Assay Kit (Thermo Fisher Scientific, Darmstadt, Germany), following the manufacturer's instructions. The protein concentration was determined by using bovine serum albumin to create a standard curve (0–1 mg·mL⁻¹) under the same experimental conditions.

For the preparative scale reaction, UMG-SP-1 was expressed in *E. coli* BL21(DE3)-Gold cultivated in terrific broth (TB) medium supplemented with 50 $\mu\text{g}\cdot\text{mL}^{-1}$ kanamycin. Auto-induction solution was added from a 50X stock to a final concentration of 1X (0.5% glycerol, 0.05% glucose, 0.2% α -lactose) and 1% (v/v) of a preculture was used to inoculate the main culture. After 4 h of incubation at 37 °C with shaking (160 rpm), the temperature was shifted to 20 °C for enzyme expression. After 16–20 h, cells were harvested by centrifugation (4,000 g, 4 °C, 20 min). Following this, cells were resuspended in 50 mM Tris-HCl, 100 mM NaCl (pH 8.0) and lysed on ice using a Sonoplus HD 2070 probe (5 min, 50% power, 50% pulse; Bandelin Electronic GmbH & Co. KG, Berlin, Germany). After centrifugation (4,000 g, 4 °C, 30 min), the clarified lysate was loaded onto a Ni-NTA column (5 mL). The column was equilibrated with 50 mM Tris-HCl, 100 mM NaCl (pH 8.0) before use. Imidazole was added to a final concentration of 20 mM to the lysate before loading it onto the column. The column was washed with 5 column volumes of washing buffer (50 mM Tris-HCl, 100 mM NaCl, 50 mM

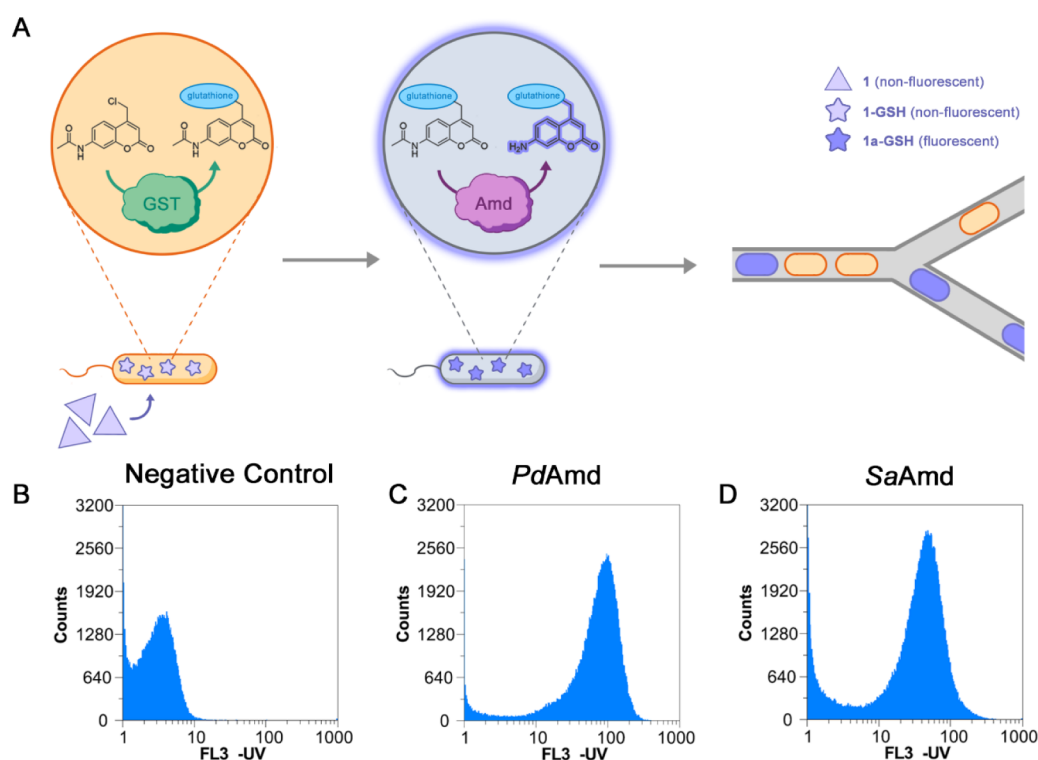


Figure 1. FACS-based detection of amidase activity. (A) Overview of the FACS-based assay principle. Fluorescence histograms of *E. coli* BL21(DE3)-Gold cells expressing GST only as a negative control (B) or co-expressing GST and the amidase PdAmd (C) or SaAmd (D), respectively. Amidase activity in whole cells towards 1 was measured by flow cytometry (FL3-UV: $\lambda_{\text{ex}} = 375 \text{ nm}/\lambda_{\text{em}} = 440 \text{ nm}$); 1 and 1-GSH do not exhibit fluorescence.

imidazole, pH 7.5). Protein was eluted on ice with elution buffer (50 mM Tris-HCl, 100 mM NaCl, 150 mM imidazole, pH 7.5) and further concentrated using a Vivaspin 20 centrifugal concentrator (10 kDa molecular weight cutoff; Sartorius, Göttingen, Germany). A PD-10 column was used to desalt the protein solution and change the buffer to 50 mM Tris-HCl, 100 mM NaCl (pH 7.5). The protein concentration was determined by measuring absorbance at 280 nm and the extinction coefficient of UMG-SP1 ($\epsilon = 1,490 \text{ M}^{-1} \text{ cm}^{-1}$).⁴³

Successful expression and purification were confirmed by 12.5% (w/v) sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE, Figures S5–6), using the Mini-PROTEAN electrophoresis system (Bio-Rad, Feldkirchen, Germany). The InstantBlueTM Protein Stain (Expedeon, Heidelberg, Germany) was used to visualize protein bands as instructed by the supplier.

Determination of Specific Activities. The commercial compounds 1-acetamido naphthalene (3, CAS No. 86-87-3), 1-aminonaphthalene (AN, CAS No. 134-32-7), *N*-(4-nitrophenyl)benzamide (4, CAS No. 3393-96-2), *p*-nitroacetanilide (5, CAS No. 104-04-1), *p*-nitroaniline (pNA, CAS No. 100-01-6), 4-methylumbelliferyl acetate (6, CAS No. 2747-05-9), 4-methylumbelliferone (UMB, CAS No. 90-33-5), and ethyl 4-nitrophenylcarbamate (8, CAS No. 2621-73-0) were used without further purification. The compound 7-acetamido-4-chloromethylcoumarin (1) was purchased from Life Chemicals, Inc. (Hohenbrunn-Riemerling, Germany). The compounds 7-carbethoxyamino-4-methylcoumarin (7, CAS No. 58632-48-7) and 7-aminomethylcoumarin (AMC, CAS No. 26093-31-2) were synthesized as described previously.⁴³ The coumarin derivative 7-acetamido-4-methylcoumarin was synthesized following the procedure by Leaym and co-

workers.⁵⁹ The following compounds were synthesized and purified as described in the SI: 1a, 1-GSH, and 1a-GSH.

Specific activities were determined as previously described.⁴³ The linear increase in fluorescence (1-GSH, 2, 3, 6, and 7) or absorbance (4, 5, and 8) was determined for 3–5 min on a VarioskanTM LUX multimode plate reader (Thermo Fisher Scientific, Darmstadt, Germany). A complete list of all determined specific activities is given in Table S3. The hydrolysis products, wavelengths, and experimentally determined molar extinction coefficients are given in Table S4.

RESULTS AND DISCUSSION

FACS-Based Assay Establishment. Prior to applying the FACS-based assay principle for the detection of amidase activity (Figure 1A), the following controls were performed. First, we showed that *E. coli* BL21(DE3)-Gold, used for the co-expression of target enzymes, is unable to cleave the amide bonds in the screening substrate 1 and the structurally related 7-acetamido-4-methylcoumarin (AAMC, 2). This proves that the host background does not interfere with the assay principle (Figures 1B and S2). Second, we confirmed that 1 and 1-GSH are not fluorescent and, finally, that the target fluorophore 1a-GSH is successfully retained within cells (Figure S3). Subsequently, the GST from *Schistosoma japonicum* (UniProtKB accession no.: P08515)⁶⁰ was solubly expressed in *E. coli* BL21(DE3)-Gold and an in-house panel of amidases was screened against 1 by FACS after preparing the corresponding co-transformants by standard transformation protocols (Figure S4). The two most suitable candidates for assay establishment purposes were found to be an amidase from *Peribacillus deserti* (PdAmd; UniProtKB accession no.: A0A2N5M658)⁶¹ and SaAmd (UniProtKB accession no.: A0A7H0LPJ3).⁶² Both

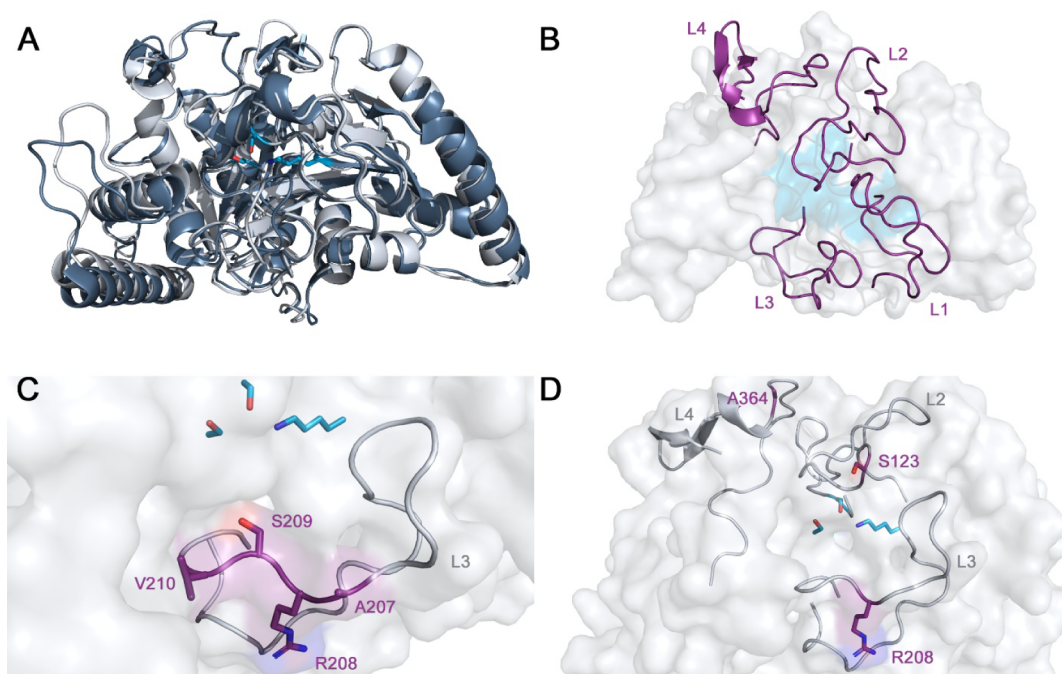


Figure 2. AlphaFold2 structural model of the amidase SaAmd. (A) Structural comparison of SaAmd (white) and the urethanase UMG-SP-3 (gray; PDB No.:9FZ1). (B) The loop regions L1–4 (purple) surrounding the active site (light blue) are highlighted. (C) Amino acid residues A207, R208, S209, and V210 in L3 (purple), targeted by saturation mutagenesis of two neighboring amino acids at a time; the catalytic triad of SaAmd is shown in blue. (D) The distal amino acid residues S123 in L2, R208 in L3, and A364 in L4 highlighted in purple, were targeted simultaneously by SSM. The SaAmd model was created using AlphaFold2.⁵⁷ Structures were visualized using PyMOL 3.0.0.

amidases belong to the AS protein superfamily³⁵ and have not been biochemically characterized before; SaAmd is a homologue to the recently characterized metagenomic urethanases UMG-SP-1, UMG-SP-2, and UMG-SP-3.^{43,44,63,64} Prior to this work, SaAmd was identified through the bioinformatic-assisted mining of the publicly accessible InterPro database (IPR036928)⁶⁵ and could be functionally annotated as a member of the AS family, capable of hydrolyzing not only amide but ester and carbamate bonds (unpublished). Cells co-expressing PdAmd and GST were incubated with **1** and showed high fluorescence signals (Figure 1C), which can be used to separate them from (non-fluorescent) *E. coli* BL21(DE3)-Gold cells expressing only GST (Figure 1B). When co-expressing SaAmd and GST, a lower signal intensity was observed (Figure 1D). The possibility to distinguish enzymes with varying activity highlights the sensitivity of the assay setup. This allows the detection and, importantly, the selection of amidases with improved activities (here, toward the screening substrate **1**), which is a hallmark of uHTS-assisted engineering campaigns.⁴

To assess the amidase activity-based separation of cells by FACS, we aimed at sorting a mixed population of cells either containing the highly active PdAmd or the less active SaAmd; the difference in activity can be inferred from a shift to higher intensities for PdAmd in the fluorescence histogram (Figure 1C). Therefore, a 1:1-mixture of cells co-expressing GST and either PdAmd or SaAmd was prepared and the top 1% of fluorescent cells were sorted (Figure S7A). Sorted cells were cultivated overnight and subsequent FACS analysis indicated an enriched population containing highly active (i.e., fluorescent) cells (Figure S7B). Cells were recovered as described above. Sanger sequencing of plasmid DNA isolated from 20 single colonies confirmed a 100% active population

containing PdAmd. Satisfyingly, this result proved the functionality of the proposed assay principle through the enrichment of a cell population containing enzymes (or variants) with enhanced amidase activity.

FACS-Assisted Engineering of Amidase Activity. Motivated by the successful establishment of our FACS-based assay, we aimed to verify its suitability through the screening of large combinatorial libraries. Therefore, we targeted SaAmd by SSM. Comparison of the primary sequence and the overall structure of SaAmd suggests homology to other AS family members, including the recently identified metagenome-derived urethanases UMG-SP-1 to 3; SaAmd shares the highest sequence identity (61.4%) with UMG-SP-3 (Figure 2A).⁴⁴ Structural elucidation of the metagenomic urethanases revealed highly flexible loops in the active site, which are important for the positioning of substrates and the overall hydrolytic activities, as supported by protein engineering and molecular dynamics simulations.^{43,63,64} Importantly, amino acid residues in these active site loops were identified as valuable engineering targets, yielding single variants with increased activity toward low-molecular weight *N*-aryl amides and carbamates.^{43,63}

Consequently, we targeted the corresponding active site loops in SaAmd by protein engineering, simultaneously randomizing two neighboring amino acid positions per library (Figure 2B). For SSM, we used primers containing double-site NNK codons (Tables S1 and S2). Successful introduction of mutations was confirmed by Sanger sequencing for all 16 desired libraries, which were screened and sorted by our FACS-assisted assay, with a gate threshold containing the top 0.5–1% of fluorescent cells (Figure S8). The most enriched mutants of each library according to Sanger sequencing were expressed and purified through C-terminal 6xHis-tags as

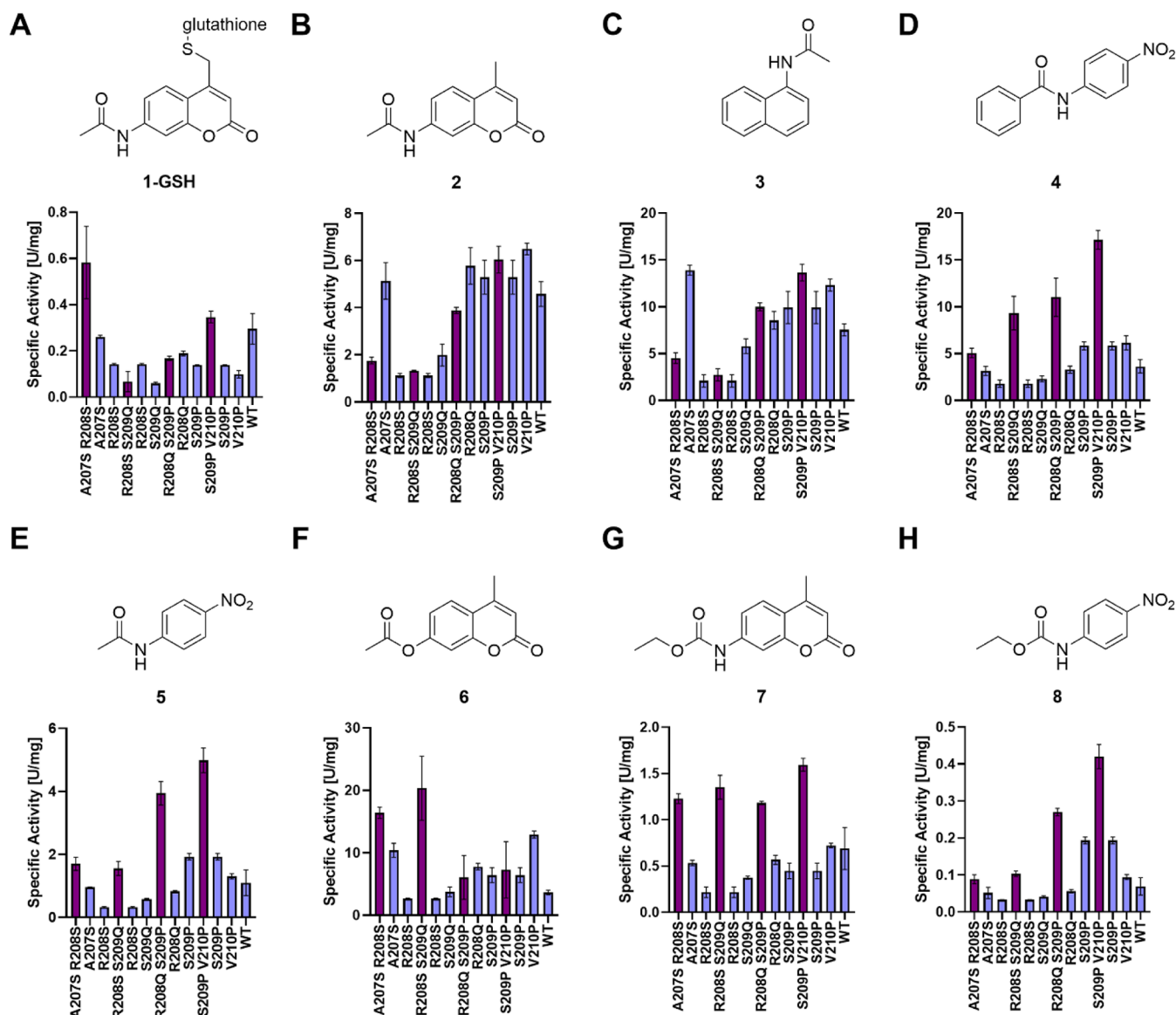


Figure 3. Specific activities of *SaAmd* and selected double-variants. Variants identified through FACS analysis are depicted in purple; respective single-variants and the wild-type enzyme are depicted in light blue. The substrate scope of enriched variants after FACS analysis was elucidated for substrates (A) 1-GSH, (B) 2, (C) AAN (3), (D) *N*-(4-nitrophenyl)benzamide (4), (E) pNAA (5), (F) 4-methylumbelliferylacetate (6), (G) CAMC (7) and (H) ENPC (8), employing standard hydrolysis conditions (100 μ M substrate in 50 mM Tris-HCl, 100 mM NaCl, 10% (v/v) DMSO; pH 7.5) at 22–24 $^{\circ}$ C as reported previously.⁴³ The increase in fluorescence (1-GSH, 2, 6, and 7: λ_{ex} = 365 nm/ λ_{em} = 440 nm; 3: λ_{ex} = 315 nm/ λ_{em} = 445 nm) or absorbance (4, 5, and 8: λ = 390 nm) from the formation of the corresponding hydrolysis products was recorded over time (0–3 min). Specific activities are shown as mean values $\pm$ standard deviation (SD) of biological replicates ($n \geq 3$).

confirmed by SDS-PAGE (Figure S6). To gain insight into potential epistatic effects of the double-variants, the respective single-variants were constructed by established site-directed mutagenesis protocols. Finally, the specific activities were determined against the GSH-conjugated 1-GSH and the structurally related coumarin derivative *N*-(4-methyl-2-oxo-2H-chromen-7-yl)acetamide (2; Figure 3A,B). The preparation and purification of both 1-GSH and the corresponding hydrolysis product 1a-GSH are described in the SI. Both substrates were converted by the wild-type *SaAmd* with specific activities of 0.30 ± 0.06 U/mg and 4.58 ± 0.48 U/mg against 1-GSH and 2, respectively (Figure 3A,B). Noteworthy, *PdAmd*, used as positive control during the establishment of the FACS-based assay, also hydrolyzed 2 (Figure S9). The most promising double-mutants obtained through FACS

screening featured mutations at the positions A207, R208, S209, and V210, located in the highly flexible active site loop L3 (Figure 2C). While the hydrolysis of 1-GSH was improved with the A207S/R208S double-variant, the corresponding single-variants (A207S and R208S) did not exhibit enhanced activity toward this substrate. R209P/V210P showed only slightly enhanced activity toward 1-GSH but hydrolyzed 2 with a higher specific activity than the wild-type *SaAmd*. For amide 2, this is also reflected by the 1.4-fold improved activity of the variant V210P over the wild-type enzyme. All other double-variants exhibited similar or lower activity against both amide substrates under experimental conditions. Based on the promiscuous hydrolytic activities of reported AS family members,^{43,44} we expanded our substrate library toward other commercially available small molecules, including the

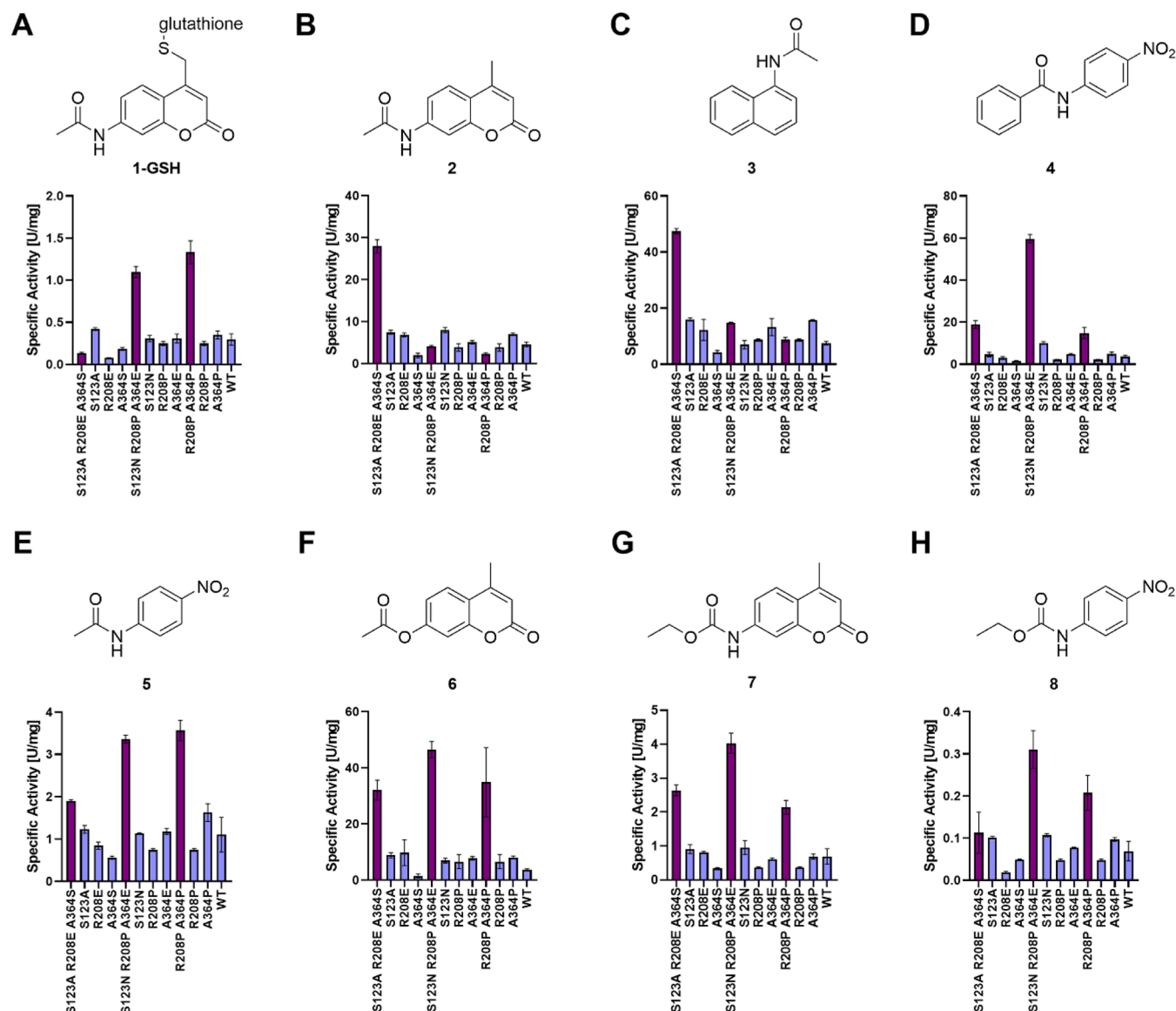


Figure 4. Specific activities of *SaAmd* and selected triple-variants. Variants identified through FACS analysis are depicted in purple. Respective single variants and the wild-type enzyme are depicted in light blue. The substrate scope of enriched variants after FACS analysis was elucidated for substrates (A) 1-GSH, (B) 2, (C) AAN (3), (D) *N*-(4-nitrophenyl)benzamide (4), (E) *p*NAA (5), (F) 4-methylumbelliferylacetate (6), (G) CAMC (7) and (H) ENPC (8), employing standard hydrolysis conditions as before. Specific activities were calculated based on the linear increase in fluorescence (1-GSH, 2, 6, and 7: $\lambda_{\text{ex}} = 365 \text{ nm}/\lambda_{\text{em}} = 440 \text{ nm}$; 3: $\lambda_{\text{ex}} = 315 \text{ nm}/\lambda_{\text{em}} = 445 \text{ nm}$) or absorbance (4, 5, and 8: $\lambda = 390 \text{ nm}$) from the formation of the corresponding hydrolysis products over time (0–3 min). Results are shown as mean specific activities $\pm$ SD of biological replicates ($n \geq 3$).

amides 1-acetamidonaphthalene (AAN, 3), *N*-(4-nitrophenyl)-benzamide (4), and *para*-nitro acetanilide (*p*NAA, 5), the structurally related ester 4-methylumbelliferylacetate (6), as well as the carbamates 7-carbethoxyamino-4-methylcoumarin (CAMC, 7) and ethyl *N*-nitrophenyl carbamate (ENPC, 8; for structures, see Figure 3).

The specific activity for 3 was improved for R208Q/S209P and S209P/V210P, whereas the double-variants A207S/R208S and R208S/S209Q showed decreased activity compared to the wild-type *SaAmd* (Figure 3C). Interestingly, the single-variant A207S exhibited similar improved activity to S209P/V210P, whereas the double-mutant A207S/R208S displayed decreased hydrolytic activity toward 3. All identified double-mutants showed improved activity against the bulky amide 4, with S209P/V210P exhibiting almost 5-fold improved activity

($17.15 \pm 0.82 \text{ U/mg}$) in comparison to the wildtype ($3.65 \pm 0.67 \text{ U/mg}$; Figure 3D). The single-variants exhibited similar or lower activity than the wild-type enzyme, suggesting that the combination of both mutations leads to the improved activity against 4. For the smaller *p*NAA (5), the double-mutants R208Q/S209P and S209P/V210P displayed enhanced activity, suggesting overall improved amidase activity conveyed by substitutions at positions S209 and V210 (Figure 3E). The single-variant S209P showed slightly improved activity, but, otherwise, the determined activities suggest that the improved activity of the double-variants is due to epistatic effects. The variants A207S/R208S and R208S/S209Q also showed greatly increased esterase activity toward 6 (up to 5.5-fold; Figure 3F). As indicated above, esterase activity has been reported for various AS family members^{44,66} and could be

confirmed for *SaAmd*. Furthermore, the hydrolysis of the small carbamate CAMC (7), which is structurally similar to the GSH-conjugated screening substrate 1-GSH and other investigated substrates (3 and 6), was significantly improved for all double-variants but not the corresponding single-variants (Figure 3G).

The variants R208Q/S209P and S209P/V210P hydrolyzed ENPC (8) more efficiently (almost 4- and 6-fold increased activity, respectively); the single-variant S209P, present in both FACS-derived double-variants, exhibited close to 3-fold improved activity (Figure 3H). Not only does this suggest that variants identified by FACS analysis show improved activity for the initial screening substrate; engineering of amino acid residues in the flexible loop L3, which is part of the active site in *SaAmd* and related AS family members, co-evolved hydrolytic activities toward other chemical bonds, including esters (Figure 3F) and highly stable *N*-aryl carbamates (Figure 3G,H).

While the simultaneous mutation of two positions with NNK codons requires the screening of 3,000 variants (including oversampling), the randomization of three positions at the same time requires the screening of 100,000 variants.²¹ To investigate the throughput of the developed FACS-based assay, we created a combinatorial library targeting the positions S123 (in L2), R208 (in L3), and position A364; the latter is located in L4 (Figure 2D). All three positions were randomized by NNK oligonucleotides. Screening, sorting and determination of specific activities were performed as described above. Amongst the identified triple-mutants, *SaAmd*-S123A/R208E/A364S (*SaAmd*-AES) showed the highest improvement in activity on 2, while the activity toward 1-GSH was reduced *in vitro* compared to the wild-type enzyme (Figure 4A,B). Contrary, the triple-variant S123N/R208P/A364E (*SaAmd*-NPE) and the double-mutant R208P/A364P (featuring the wild-type serine at position 123; *SaAmd*-SPP) show higher conversion of 1-GSH but do not exhibit improved activity for 2. These enhanced activities are not observed in the single-variants, hinting toward synergistic effects exerted by the combination of mutations. Cleavage of 3 was most significantly improved for *SaAmd*-AES (47.44 ± 0.89 U/mg), followed by *SaAmd*-NPE (14.75 ± 0.19 U/mg); *SaAmd*-SPP showed activity against 3 similar to the wild-type *SaAmd* (7.53 ± 0.6 U/mg; Figure 4C). While triple-variants *SaAmd*-AES and *SaAmd*-SPP showed improved activity for the benzamide derivative (4) similar to the investigated double-mutants, *SaAmd*-NPE exhibited the highest specific activity observed against 4 (59.7 ± 1.75 U/mg), which is a 16.3-fold improvement over the wild-type enzyme (3.65 ± 0.67 ; Figure 4D). Hydrolysis of *p*NAA (5) was improved 3-fold for *SaAmd*-NPE (3.36 ± 0.07 U/mg) and 3.2-fold for *SaAmd*-SPP (3.56 ± 0.21 U/mg; Figure 4E). *SaAmd*-NPE displayed an 11.7-fold enhanced hydrolysis of the coumarin-derived ester 6 (46.48 ± 2.58 U/mg; Figure 4F) compared to the wildtype (3.66 ± 0.34 U/mg). Interestingly, all three triple-variants exhibited improved activities toward the carbamate 7, with *SaAmd*-NPE displaying the highest observed fold-increase of 5.9 (Figure 4G). Hydrolytic activity against the smaller carbamate ENPC (8) by *SaAmd*-SPP and *SaAmd*-NPE was higher than the wild-type enzyme, but only in the range of the double-variants obtained from the previous screening of L3 featuring neighboring mutations (Figure 4H, see also Figure 3). The majority of these single-variants did not show a significant increase in activity against most substrates. However, S123A

and exchanges at position A364 (A364P and A364E) yielded variants with increased activity against certain substrates (Figure 4). Particularly A364P exhibited overall enhanced activity against different substrates, including the stable carbamate 8; CAMC (7) was hydrolyzed with a specific activity comparable to the wildtype. Due to the fact that the majority of single-mutants did not exhibit increased activity, their combination would not have been obvious based on SSM of individual sites and (low-throughput) biochemical characterization. This also emphasizes non-additive effects along activity improvements, which might even be harder to predict since the targeted amino acid residues are located in different, highly movable loop regions. Thus, we demonstrate the great value of our FACS-based assay by identifying distant but in their combination beneficial mutations, providing *SaAmd* variants – AES, NPE, and SPP – with significantly enhanced hydrolytic profiles. Especially in the case of the triple-variants, no indication for a beneficial combinatorial effect of the respective single-mutants was given based on specific activities, suggesting epistatic interactions that contribute to enhanced hydrolytic profiles of *SaAmd* variants as observed with other engineering targets.^{16–18,67,68}

CONCLUSION AND OUTLOOK

We successfully established a flow cytometry-based uHTS assay for the assessment of amidase activity in whole cells. The FACS setup was suitable to distinguish AS family members with varying activity (*PdAmd* and *SaAmd*) for the coumarin-derived screening substrate 1; sorting yielded a population exclusively containing cells expressing the more active *PdAmd*. The reported whole-cell assay system allows for an easy linkage between genotype and phenotype through the GST-catalyzed modification of the coumarin-derived screening substrate, which could be applied to other reporter molecules as well. Employing well-established protocols for the co-expression of enzymes in commonly used *E. coli* BL21(DE3)-Gold cells promises the system to be easily transferable to other laboratories and allows the exchange of strains and/or compatible plasmids if desired. By replacing the amide group of the FACS screening substrate by another desired functional group, the application of our assay can certainly be extended toward other enzyme classes and, thereby, adapted to other enzymatic reactions of interest.^{28,69}

Importantly, the applicability of our FACS-assisted assay was demonstrated by screening large combinatorial libraries of up to three simultaneously randomized positions in active site loops of the so far uncharacterized *SaAmd* enzyme. Identified variants exhibited up to 16.3-fold improved hydrolytic activity on different amide substrates, therefore, demonstrating that the results achieved with the GSH-conjugated surrogate substrate can be reliably transferred to other small molecules containing amide bonds. Conveniently, co-evolution for the hydrolytic activity of *SaAmd* toward ester and carbamate bonds was achieved, as (purified) variants exhibited up to 11.7-fold improved activity. This is of significance since urethanases, acting on notoriously stable *N*-aryl carbamates, have only been discovered very recently,^{43,44} and tools to accelerate their engineering are highly desired. Together, this not only expands the substrate scope of *SaAmd*; FACS-derived variants showed improved hydrolytic profiles toward other small molecules – even beyond the employed screening substrate. Furthermore, the developed FACS-assisted assay is applicable for the screening of large libraries, constructed by epPCR, SSM,

other mutagenesis techniques, or even derived from metagenomes. This facilitates the identification of the desired enzymatic activity and enables the evaluation of combinatorial effects of proximal as well as distal mutations. Such epistatic effects are otherwise difficult to predict and can be easily missed by established activity-informed combination of single-mutants from low-throughput engineering campaigns. Therefore, we not only advance the repertoire of uHTS strategies for amidases as demonstrated but provide a highly customizable FACS-based screening platform for the tailoring of biocatalysts. The enhanced activities of newly characterized SaAmd variants and their structurally relatedness to metagenomic urethanases make them promising biocatalysts for biotechnological applications in the recycling of synthetic polymers like polyamides and polyurethanes, which will be investigated in the future.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.5c01903>.

Additional figures and tables, details on experimental procedures, and NMR spectra (PDF)

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Funding

A.C.C. and T.G. were supported by the Austrian Science Fund (FWF) Project P36101B (Grant-DOI: [10.55776/P36101](https://doi.org/10.55776/P36101)).

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors would like to thank P. Hildebrandt for assistance with initial flow cytometry experiments and D. Böttcher for fruitful discussion. We also thank I. Faber and I. Menyes for resource management.

■ ABBREVIATIONS

AACMC	7-acetamido-4-chloromethylcoumarin
AAMC	7-acetamido-4-methylcoumarin
AAN,	1-acetamidonaphthalene
AMC	7-aminomethylcoumarin
AN	1-aminonaphthalene
AS	amidase signature
CAMC	7-carbethoxyamino-4-methylcoumarin
CFE	cell-free extract
DMSO	dimethyl sulfoxide
ENPC	ethyl N-nitrophenyl carbamate
epPCR	error-prone PCR
<i>E. coli</i>	<i>Escherichia coli</i>
FACS	fluorescence-activated cell sorting;
GSH	glutathione
GST	glutathione S-transferase
LB	lysogeny broth
PCR	polymerase chain reaction
pNA	para-nitroaniline
pNAA	para-nitro acetanilide
SDS-PAGE	sodium dodecyl sulfate-polyacrylamide gel electrophoresis
SSM	site-saturation mutagenesis

TB terrific broth
uHTS ultrahigh-throughput screening
UMB 4-methylumbelliferone

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